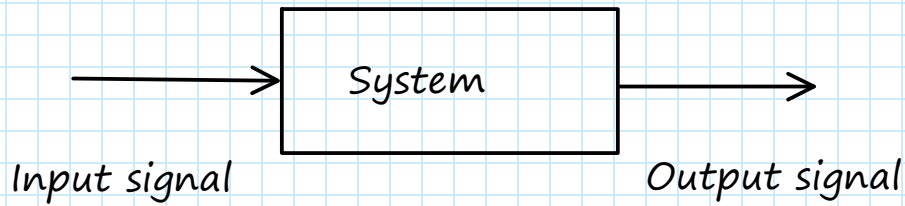


Lctr 2: DT signals and systems



Learning outcomes:

- LO2. Express signals in terms of basic signals
- LO3. Determine periodicities of DT signals
- LO4. Determine linearity, time invariance, causality, and stability of DT systems

LO2. Express signals in terms of basic signals

- Def: DT signal/sequence x

$$x: \text{Integers} \rightarrow \text{Complex}$$

DT signal x is a function that maps integers to complex values

$x[n]$: the value of x at index n .

$\{x[n]\}$: the collection of all values of x .

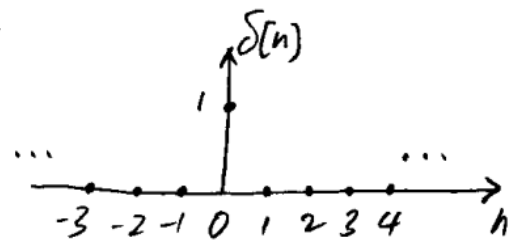
We usually say $x[n]$ for both the entire signal/sequence and the value at index n .

- Basic signals

- The DT impulse of unit sample

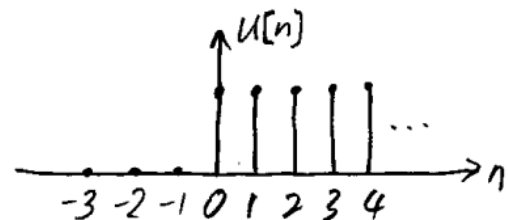
$$\delta[n] = \begin{cases} 1, & n=0 \\ 0, & \text{otherwise} \end{cases}$$

$$= u[n] - u[n-1]$$



- The DT unit step

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$



$$= \delta[n] + \delta[n-1] + \delta[n-2] + \dots$$

$$= \sum_{k=0}^{\infty} \delta[n-k]$$

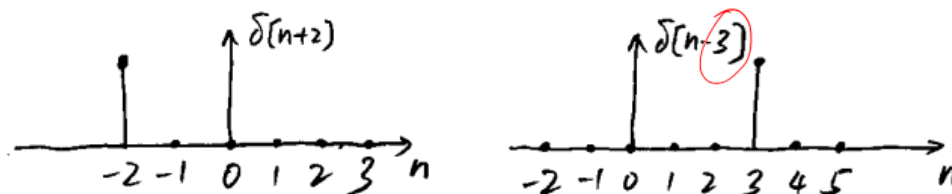
- Basic systems

- Shift



$n_0 > 0$: delay by n_0 ticks (shift right)

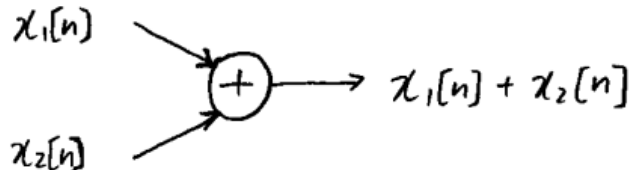
$n_0 < 0$: advance by $-n_0$ ticks (shift left)



- Scale



- Add

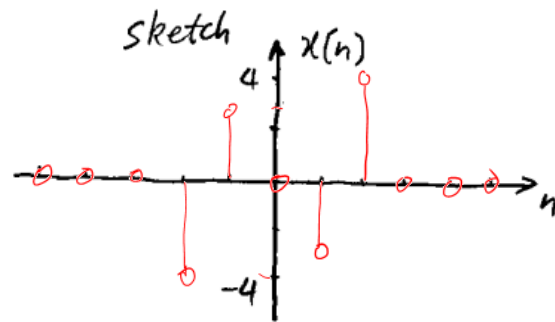


- Other signals

- Any DT signal can be represented as a sum of scaled and delayed unit impulses

Example:

$$x[n] = \begin{cases} -4, & n = -2 \\ 3, & n = -1 \\ -3, & n = 1 \\ 4, & n = 2 \\ 0, & \text{otherwise} \end{cases}$$



Formula:

$$x[n] = -4\delta[n+2] + 3\delta[n+1] - 3\delta[n-1] + 4\delta[n-2]$$

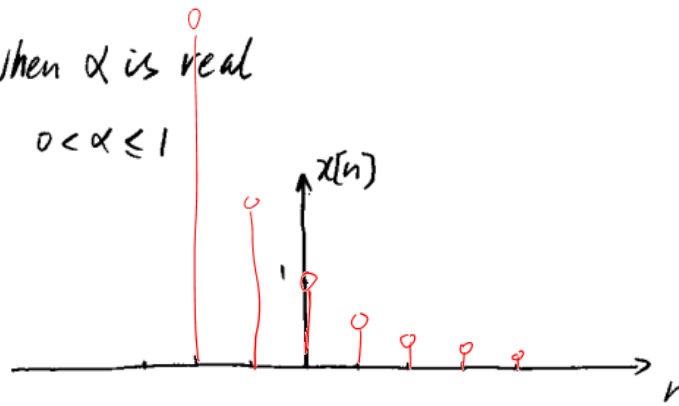
□

• Exponential sequences

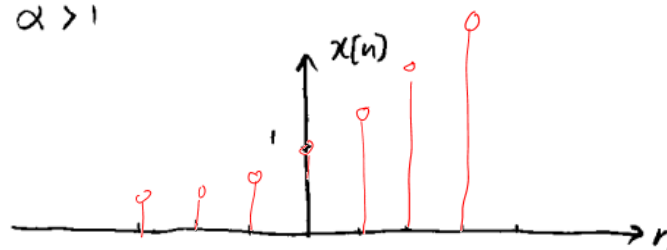
$$x[n] = \alpha^n$$

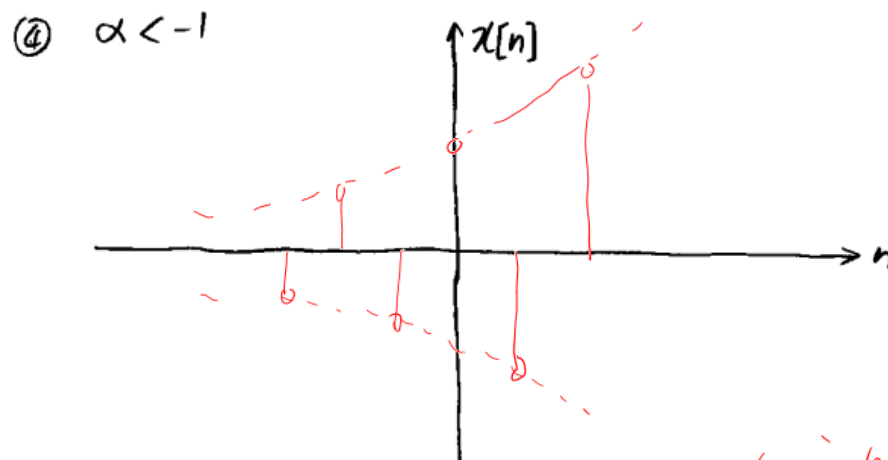
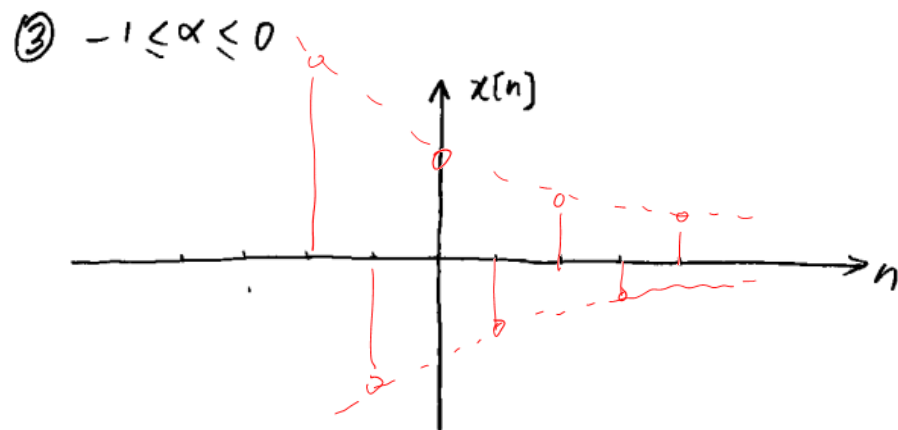
— When α is real

① $0 < \alpha \leq 1$



② $\alpha > 1$





— When α is complex

$$\alpha = |\alpha| e^{j\omega_0}$$

$$\rightarrow x[n] = \alpha^n = |\alpha|^n e^{j\omega_0 n}$$

$$= |\alpha|^n (\cos(\omega_0 n) + j \sin(\omega_0 n)).$$

$$(e^{j\omega_0})^n = e^{j\omega_0 n}$$

LO3. Determine periodicities of DT signals

- Def: A DT signal $x[n]$ is said to be periodic with period N when

$$x[n+N] = x[n], \quad \forall n \in \mathbb{I}, \quad N \in \mathbb{I}$$

- Question:

For what values of ω_0 can $e^{j\omega_0 n}$ be periodic with period $N \in \mathbb{I}$?

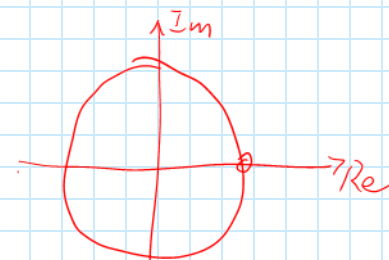
Soln: $e^{j\omega_0(n+N)} = e^{j\omega_0 n} \Rightarrow e^{j\omega_0 n} \cdot e^{j\omega_0 N} = e^{j\omega_0 n}$

$$\Rightarrow e^{j\omega_0 N} = 1$$

$$\Rightarrow \omega_0 N = 2\pi k$$

$$\Rightarrow \omega_0 = 2\pi \cdot \frac{k}{N}$$

only need to find a single k and N .



Result: A DT sinusoid or complex exponential is periodic with period N when $\exists k \in \mathbb{I}$ such that

$$N = \frac{2\pi}{\omega_0} k \in \mathbb{I}$$

- Def: The smallest value of the above N is the fundamental period.

Example: Is $e^{j\frac{3\pi}{4}n}$ periodic? If yes, what is the fundamental period?

Soln:

$$(a) \quad e^{j\frac{3\pi}{4}(n+N)} = e^{j\frac{3\pi}{4}n}$$

requires $e^{j\frac{3\pi}{4}N} = 1$

which requires $\frac{3\pi}{4}N = 2\pi k$

which can be satisfied if $\frac{3N}{8} = \frac{2k}{3}$

$\Rightarrow e^{j\frac{3\pi}{4}n}$ is periodic with $N=8$

$$(b) \quad \frac{3\pi}{4}N = 2\pi k$$

$\Rightarrow N = \frac{8}{3}k$

For $k=3$, $N=8$, which is the fundamental period.

□

$$\frac{\pi}{5}N = 2\pi k \quad N=10k$$

Question: Is $e^{j\frac{\pi}{5}n}$ periodic? Period?

10

Is e^{jn} periodic? period?

$N=2\pi k$ X

LO4. Determine linearity, time-invariance, causality, and stability of DT systems

$$x[n] \longrightarrow \boxed{T\{\}} \longrightarrow y[n] = T\{x[n]\}$$

$T\{\}$ is the rule by which $y[n]$ is formed using $x[n]$

- Linearity: $T\{\}$ is linear iff

$$T\{a x_1[n] + b x_2[n]\} = a T\{x_1[n]\} + b T\{x_2[n]\}$$

- Remarks: To prove/disprove linearity, use the rule to see what the output is for each of $x_1[n]$, $x_2[n]$, and $a x_1[n] + b x_2[n]$ as input; then see if the above linearity condition is observed.

- Example: Is $y[n] = T\{x[n]\} = \frac{1}{x[n+1]}$ linear?

$$\text{Soln: } x[n] = x_1[n] \Rightarrow y[n] = y_1[n] = \frac{1}{x_1[n+1]}$$

$$x[n] = x_2[n] \Rightarrow y[n] = y_2[n] = \frac{1}{x_2[n+1]}$$

$$x[n] = x_3[n] = a x_1[n] + b x_2[n]$$

$$\Rightarrow y[n] = y_3[n] = \frac{1}{a x_1[n+1] + b x_2[n+1]}$$

$$y_3[n] \neq a y_1[n] + b y_2[n]$$

$$\Rightarrow T\{\} \text{ is not linear.}$$

□

- Time invariance: $T\{\}$ is time invariant iff

$$T\{x[n-n_0]\} = T\{x[n]\} \text{ shifted by } n_0.$$

- Remarks: The same basic approach as above, but only need one test: the output of the delay input vs the delayed output of the input.

- Example: Is $y[n] = T\{x[n]\} = \frac{1}{x[n+1]}$ time invariant?

$$\begin{aligned} \text{Soln: } x[n] = x_1[n] &\Rightarrow y[n] = y_1[n] = \frac{1}{x_1[n+1]} \\ &= y_1[n-n_0] = \frac{1}{x_1[n-n_0+1]} \end{aligned}$$

$$\begin{aligned} x[n] = x_1[n-n_0] = x_2[n] &\Rightarrow \\ y[n] = y_2[n] &= \frac{1}{x_1[n-n_0+1]} \end{aligned}$$

$\Rightarrow$ Time invariant.

□

- Causality: $T\{\}$ is causal iff $y[n] = T\{x[n]\}$ is independent of future values of $x[n]$; i.e., $y[n]$ only depends on

$$x[n], x[n-1], x[n-2], \dots,$$

But not

$$x[n+1], x[n+2], \dots$$

- Example: Is $y[n] = T\{x[n]\} = \frac{1}{x[n+1]}$ causal?

Soln:

□

- BIBO (Bounded-input bounded-output) stability:
 $T\{\}$ is BIBO stable iff

$$|x[n]| \leq M_x < \infty \Rightarrow |y[n]| \leq M_y < \infty, \forall n \in \mathbb{Z}.$$

Example: Is $y[n] = T\{x[n]\} = \frac{1}{x[n+1]}$ BIBO stable?

Soln:

Example: Is $y[n] = T\{x[n]\} = \cos(\omega_0 n) \cdot x[n]$

Linear, time invariant, causal, and BIBO stable?

Soln: